

DESEMSETTI TEJA DURGA SAI

Lakshmaneswaram, Narasapur, W.G, A.P
+91 9010730497 | tejadesemsetti999@gmail.com

 <https://gen-ai-profile.vercel.app/>

 <https://www.linkedin.com/in/teja-desemsetti/>

Professional Summary

Generative AI Engineer with 4+ years of professional experience, including expertise in building AI-powered applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, FastAPI, and Vector Databases. Skilled in developing scalable AI systems, REST APIs, prompt engineering, embedding pipelines, and full-stack applications using React and Python. Strong experience in deploying production-grade GenAI solutions with improved response accuracy and optimized retrieval performance.

Experience

- DevZen Software Solutions Pvt Ltd** October-2022 - Present
GenAI Engineer
 - Designed and developed AI-powered applications using LLMs and RAG architecture for domain-specific business use cases.
 - Built scalable document retrieval and embedding pipelines using LangChain, improving response relevance by 35%.
 - Integrated Mistral AI and Hugging Face Transformer models for optimized inference and intelligent response generation.
 - Developed RESTful APIs using FastAPI and Flask for seamless AI model integration with frontend systems.
 - Implemented prompt engineering techniques that reduced hallucinations by 25% and improved contextual accuracy by 40%.
 - Worked with vector databases and MongoDB Atlas for efficient semantic search and retrieval operations.
 - Collaborated with cross-functional teams to deploy scalable AI solutions into production environments.
 - Improved application performance, response time, and scalability for enterprise AI applications.

Project

- Hybrid Conversational RAG Chatbot**
 - Developed an intelligent AI chatbot using Retrieval-Augmented Generation (RAG) architecture.
 - Built document ingestion, chunking, embedding, and retrieval pipelines using LangChain.
 - Integrated Mistral models and Hugging Face embeddings for high-quality contextual responses.
 - Utilized MongoDB Atlas and vector storage for efficient similarity search operations.
 - Applied advanced prompt engineering techniques to improve response accuracy and reduce hallucinations.
 - Developed REST APIs and integrated with React.js frontend for seamless interaction.
 - Achieved significant improvement in contextual relevance for real-world user queries.

- **AI Agent Automation Platform**

- Developed AI agents using LangChain for autonomous task execution.
- Built retrieval workflows integrating vector databases.
- Explored GraphRAG concepts for knowledge graph-based retrieval.
- Experimented with LoRA fine-tuning techniques for domain-specific adaptation.

- **Employee Payroll Application**

- Developed a full-stack payroll management application for automating employee salary processing.
- Built responsive frontend dashboards using React.js for HR and admin operations.
- Implemented backend services for payroll calculations, employee management, and reporting.
- Integrated secure authentication and role-based access control mechanisms.
- Designed and managed MySQL database schemas for scalable data management.

Skills

- Experience working in agile development environments
- Strong problem-solving and analytical skills
- Passionate about AI innovation and scalable system design
- Excellent Communication Skills

Technical Skills

- **Languages:** Python, JavaScript
- **Frontend:** React.js, Streamlit
- **Backend:** Python (FastAPI / Flask)
- **GenAI & ML:** LangChain, LLMs, RAG, Prompt Engineering, Fine Tuning, Hugging Face Transformers, OpenAI API
- **Databases:** MongoDB Atlas, Pinecone, MySQL
- **Tools & Platforms:** Git, REST APIs, Docker, Vector Databases
- **Cloud & DevOps:** AWS Basics, CI/CD Pipeline, Scalable Architecture

Education

- **Swarnandhra college of Engineering and Technology** 2018 - 2022
Bachelor of Technology (ECE)
CGPA : 6.8

Course And Certificates

- Generative AI / Machine Learning Certification
- HTML, CSS, JAVASCRIPT
- AWS cloud practitioner
- AI Tools Workshop